

Group Assignment 2
Role-Play Activity: Public Engagement with Nuclear Power

Background

The world's first nuclear power plant supplying electricity to a grid was commissioned in the Soviet Union in 1954, followed by India in 1969. As Ramana and Saini highlight, [nuclear power operates on complex scientific principles](#). Initially hailed as a source of cheap and limitless energy, major accidents—such as Three Mile Island (1979), Chernobyl (1986), and Fukushima (2011)—eroded public trust. In the aftermath of Fukushima, several nations, including Germany and Italy, halted or [scaled back their nuclear programs](#).

However, in recent years, the urgency of climate change, and the growing energy demands of AI, cloud computing, and electric vehicles have [reignited interest in nuclear power](#). Governments, corporations, and scientists are increasingly advocating for nuclear energy as a viable solution to meet these challenges.

Activity Objective

This role-play exercise encourages students to critically examine how public perspectives can shape complex science and technology (S&T) decisions, particularly concerning nuclear power. By engaging with diverse viewpoints on risk and uncertainty, you will explore the democratic dimensions of such policy debates.

Role Play Case: Proposed Kovvada Nuclear Power Project

Project Overview

The Government of India plans to establish a large-scale nuclear power plant in **Kovvada, Srikakulam district (Andhra Pradesh)**, consisting of **six reactors** with a combined capacity of **10,000 MW**. This project is designed to meet the long-term (50-year) energy demands of **Andhra Pradesh and Telangana** while supporting India's transition to **low-carbon electricity generation**.

Fuel Supply & Supporting Infrastructure

To ensure a **steady uranium supply** for the reactors, the government is also developing:

1. Uranium Mining & Processing Plant in Tummalapalle (Kadapa district, Andhra Pradesh).

2. Additional Uranium Mining Site in Lambapur-Peddagattu (Nalgonda district, Telangana).

These facilities aim to **reduce dependency on imported nuclear fuel** and strengthen India's self-reliance in nuclear energy.

Key Objectives

- Provide **clean, baseload electricity** to support industrial and domestic needs.
- Reduce **greenhouse gas emissions** compared to fossil fuel-based power.
- Boost **regional economic development** through infrastructure and employment.

Group Division

The class is divided into two broad cohorts:

1. **Proponents of Nuclear Power** – Experts, policymakers, and industry advocates supporting nuclear energy.
2. **Skeptics/Opponents of Nuclear Power** – Environmentalists, concerned citizens, and activists questioning its safety and sustainability.

Each cohort is further split into subgroups of 2-4 students, each representing a specific stakeholder perspective (e.g., government officials, scientists, environmental NGOs, or local communities).

Role-Play Groups: Nuclear Power Debate

Cohort 1: Proponents of Nuclear Power

1. Nuclear Power Corporation of India (NPCIL)
2. Government of India
3. Government of Andhra Pradesh
4. Atomic Energy Regulatory Board (AERB) – Regulatory & Safety Oversight
5. Bhabha Atomic Research Centre (BARC) – Research & Development Scientists
6. Government of Telangana
7. Andhra Pradesh Power Generation Corporation Limited (APGENCO)
8. ROSATOM (Russian State Nuclear Energy Corporation) – International Partner
9. Uranium Corporation of India Ltd (UCIL) + Atomic Minerals Directorate (AMD) – Fuel Supply & Exploration

10. Westinghouse Nuclear – Private Sector Technology Provider

Cohort 2: Skeptics/Opponents of Nuclear Power

11. National Alliance of People's Movements (NAPM) (<https://napmindia.org>) – Grassroots Anti-Nuclear Advocacy

12. DIANuke (<https://www.dianuke.org>) – Anti-Nuclear Information Hub

13. Save Nallamala Forest Campaign – Environmental & Biodiversity Concerns

14. Kovvada Region Residents & Farmers Collective – Local Community Opposition

15. Human Rights Forum (<https://humanrightsforum.org>) – Civil Rights & Justice Advocacy

16. Class Media Team – Task: Recording the Session (9th April)

17. Lawyers & Scientists Collective – Legal & Technical Support for Public

18. Anuvidyut Kendram Vyatireka Porata Udyamam (AKVPU) – Local Anti-Nuclear Struggle

19. Chenchu Tribe Collective – Indigenous Rights & Land Protection

20. People's Movement Against Nuclear Energy (PMANE) – Mass Mobilization Against Nuclear Projects

Preparation Tasks

- **Research:** Subgroups must review assigned readings, documentaries, and region-specific case studies to build their arguments.
- **Materials:** Prepare talking points, posters/placards, and short audio-visual aids (if needed) to strengthen your presentation.
- **Role Adherence:** Regardless of personal views, you must convincingly represent your assigned group's stance.

Presentation Guidelines

- Each subgroup will have **4 minutes** to present their case.
- You may dress and acquire manners that are appropriate for the group you are representing.
- Use visual aids (posters, whiteboards) to highlight slogans, statistics, or key arguments.
- You may create video presentations and work with other relevant sub-groups in your cohort to make your presentations impactful.

- Be persuasive, fact-based, and engaging—this is a simulation of real-world policy debates!

Assigned Readings and Suggested Resources (Please do refer to additional resources apart from these for your work):

Crownhart Casey (2025). What's next for nuclear power. The MIT Technology Review.
<https://www.technologyreview.com/2025/01/14/1109963/whats-next-for-nuclear-power/>

Collins, Harry & Pinch Trevor (1998). The science of the lambs: Chernobyl and the Cumbrian sheepfarmers (pp. 1029-1031), in *The Golem at Large: What You Should Know About Technology*, 2nd edition, Cambridge University Press: Cambridge.

Davidson, Kief (Director) (2022). *Meltdown: Three Mile Island* [documentary]. Netflix

Khan, Ajmal (2023). *People Against Nuclear Energy: Antinuclear Movements in India*, SAGE: New Delhi.

Nishiura, Masaki & Nakata, Hideo (Directors) (2023). *The Days* [miniseries]. Netflix

Ramana, M. & Saini, Sajan (2017). How do nuclear power plants work? TED-Ed.
<https://www.youtube.com/watch?v=R7WPEYGr1Vs>

Renck, Johan (Director) (2019). *Chernobyl* [miniseries]. HBO.

Sterling, Andy (2010). Keep it complex. *Nature* 468(7327): 1029-1031

National Geographic. *Unlocking Power Tarapur NPP*.
<https://www.youtube.com/watch?v=Lv1Rwj8950g>